Anna Sofia Moro

MSc Student in Physics

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Education

2023 – Present **M.Sc. Physics** Università degli Studi di Padova. *(Padua, Italy)*
Thesis title: *Characterization of Long Bent Silicon Crystals for Channeling Applications at the LHC.*

2020 – 2023 **B.Sc. Physics** Università degli Studi di Padova. *(Padua, Italy)*
Thesis title: *Performance of the Final ENUBET Demonstrator Using CERN-PS Beams.*

Projects & Research Activities

Jul. 2026 - Dec. 2026 **Short Term Internship - CERN** (Geneva, Switzerland).
Title: *Characterization of Long Bent Silicon Crystals for Channeling Applications at the LHC.*
Description: Evaluated the performance of bent silicon crystals for large-angle hadron deflection. Analyzed experimental beam test data. Benchmarked simulation models against experimental results to assess crystal viability for future LHC fixed-target setups.

Mar. 2024 - Mar. 2025 **Advanced Physics Laboratory - Università degli Studi di Padova**.
Title: *Rutherford scattering measurements.*
Description: Quantified Rutherford scattering cross-sections. Designed and implemented a Monte Carlo simulation to model the apparatus, optimize experimental parameters, and benchmark theoretical expectations.

Jun. 2024 - Oct. 2024 **Introduction to Research - Università degli Studi di Padova**.
Title: *Optimization studies for the measurement of* $\mathrm{BR}(\Lambda_b^0 \to \Lambda_c^+ D_s^{(*)-})$.
Description: Optimized the branching fraction measurement of the $\Lambda_b \to \Lambda_c^* D_s^-$ decay using LHCb data to constrain irreducible backgrounds in Lepton Flavour Universality tests.

Sep. 2023 - Dec. 2023 **Bachelor's Thesis - Università degli Studi di Padova**.
Title: *Performance of the Final ENUBET Demonstrator Using CERN-PS Beams.*
Description: Analyzed data collected at the CERN-PS for the ENUBET experiment to develop a novel calibration method independent of reconstructed silicon tracks.

Skills

Technical

Programming Python, C/C++, R
Physics Tools ROOT, RooFit, Geant4, scikit-learn, NumPy, Pandas
Tools & OS Git, LaTeX, Linux/Bash, MS Office

Language

Italian Native speaker
English C1
French A2
German A1